

Figure 6: Docker adoption trends as reported by Datadog

Resource utilisation in Docker containers is maximised and isolation leads to better and more predictable system performance. Moreover, security is enhanced; even if a user is able to compromise the system and gain superuser privileges, this user is isolated from the rest of the system. Portability to many operating systems and the version control feature make the technology an attractive package for widespread adoption across the industry.

Rapid adoption

Docker has fast become the industrial standard for containerisation, with a 40 per cent increase in adoption across all hosts, claimed a study published by the leading metrics and monitoring provider, Datadog. The study further claimed that there is a linear increase in the containers deployed by users, with the numbers nearly quintupling in the first 10 months of being introduced to containers. The major trend observed has been the deployment of several containers simultaneously on a single host.

In a similar study, leading analytics provider Sysdig claims to have observed as many as 95 containers running on a host. A majority of the hosts used Kubernetes as an orchestrator to manage these containers, while maintaining a registry to enable easy sharing between users and services.

The future

An analysis of current trends

The DockerCon 2016 keynote speech by Docker's CEO Ben Golub provided many insights into the current level adoption of this technology:

- Dockerised applications have grown about 3100 per cent over the past two years to about 460,000 in number.
- Over 4 billion containers have been shared among the developer community. Analysts are touting 8 billion as the number for 2017.
- There are nearly 125,000 Docker Meetup members worldwide, which is nearly half of the population of Iceland.

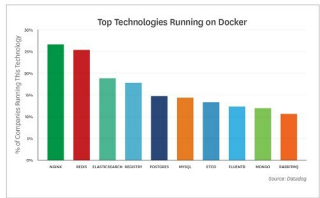


Figure 7: Top technologies running on Docker

Docker has spurred the development of a number of projects including CoreOS and Red Hat's Project Atomic, which are designed to be minimalist environments for running containers. The top 10 technologies running in Docker environments, as listed by Datadog, are shown in Figure 7.

Projections

What began with 1146 lines of code has today turned into a billion-dollar product. Docker has grown to a stage where a majority of leading tech firms have been motivated into releasing additional support for deploying containers within their products. Examples include Amazon integrating Docker into the Elastic Beanstalk system, Google introducing Docker-enabled 'managed virtual machines', and announcements from IBM and Microsoft with regard to Kubernetes support for multi-container environments. Docker has already become a part of major Linux distributions such as Ubuntu, CentOS and Red Hat Enterprise Linux (RHEL), although the packaged versions often fail to keep up with the latest releases.

The Docker team has clearly laid out its goals of developing the core capabilities, cross-service management and messaging. It is likely that, in the future, there will be more focus on building and deploying rather than the level of virtualisation, which is when the fine line between VMs and containers becomes fuzzy.

In any case, the fact that Docker has breached the billion-dollar evaluation mark is an indication of the underlying potential of this technology. Docker is clearly here to stay! 

By: Swapneel Mehta

The author has worked with Microsoft Research, CERN and startups in AI and Cybersecurity. An open-source enthusiast, he enjoys spending his time organising software development workshops for school and college students. You can connect with him at <https://www.linkedin.com/in/swapneelm> and find out more at <https://github.com/SwapneelM>.